User Management Fundamentals In Linux

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Course Name : INT305 Network Security and Protocols

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Date of Submission : 04/02/2026
Version : 1.0

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Executive Summary

This lab focused on exploring user management on the Linux operating system. Key activities include creating a user account and modifying the password, locking and unlocking the user account, terminating the session of the user account and deleting the user account. The lab revealed that created users have limited permission and do not have a super user privilege by default. A super user retains the permission and privileges to modify the operational access of the created users

1.0 Introduction

User management in Linux refers to the administration of user accounts and access control within a multi-user operating system. It involves creating, modifying, and deleting user accounts, assigning users to groups, managing passwords, and defining permission levels to regulate access to system resources. Each user in Linux is associated with a unique username, User ID (UID), group membership, and a home directory, which together help maintain organization and security. Through proper user management, administrators can enforce the principle of least privilege, ensuring that users only have access to the files and commands necessary for their roles, thereby enhancing overall system stability and protection against unauthorized access.

2.0 Aim and objectives

The aim of this lab is to learn how to create, modify and manage user accounts securely in Linux.

The Objectives of this lab is to

- create user account and modify the password
- lock and unlock user account
- terminate the session of user account
- delete user account

2.1 Tools

- Kali Linux

3.0 Methodology

3.1 Creating a User Account

- Command **`: sudo adduser student1`**
- purpose : Create the user (interactive: set password, home directory.)

- Command **`: id student1`**

- purpose : Verify the user was created

- Command **: `sudo passwd student1`**
- Purpose : Change/set a secure password for the user

- Command **: `sudo groupadd students`**
- Purpose: Create the group

- Command **: `sudo usermod -aG students student1`**
- Purpose : Add student1 to the students group (use -a to append)

- Command **: `id student1`**
- Purpose : Verify group membership

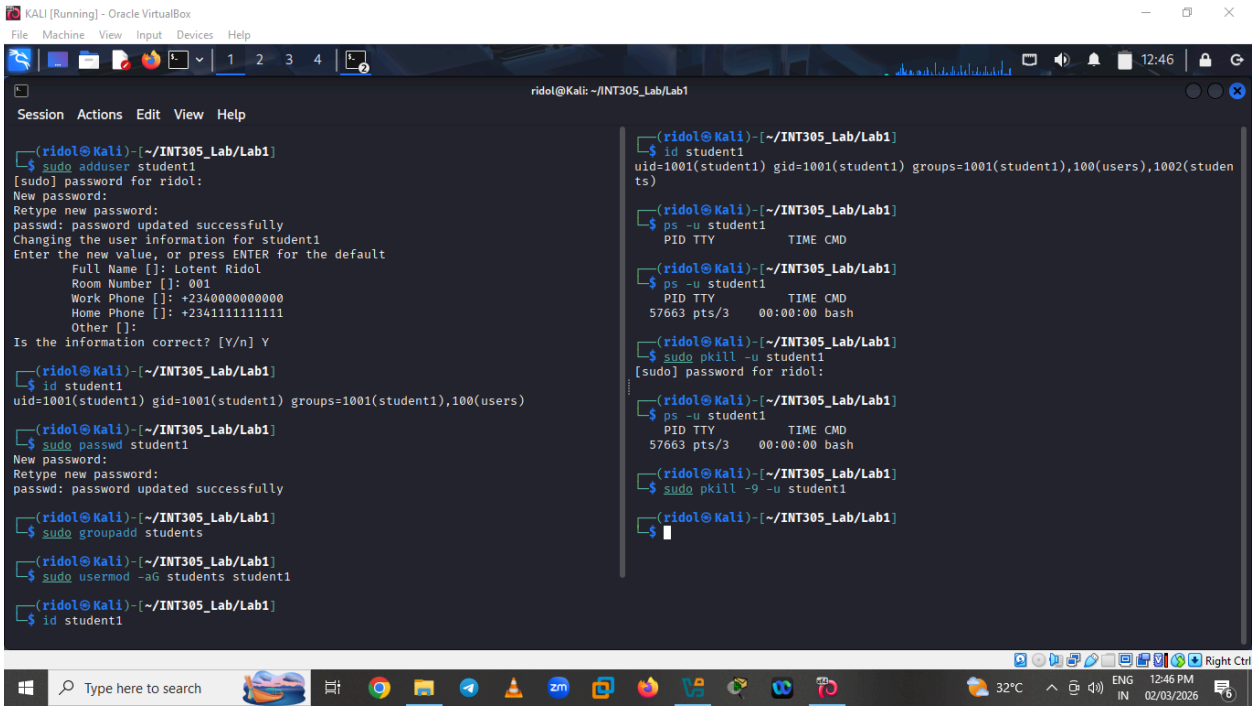

Fig 1: Evidence showing a new user being created on the Linux terminal

3.2 Terminating the session of a user

- Command : **`ps -u student1`**
- Purpose : Check if student1 has any running processes

- Command : **`sudo pkill -u student1`**
- Command : **`sudo pkill -9 -u student1`**
- Purpose : Kill all processes owned by student1

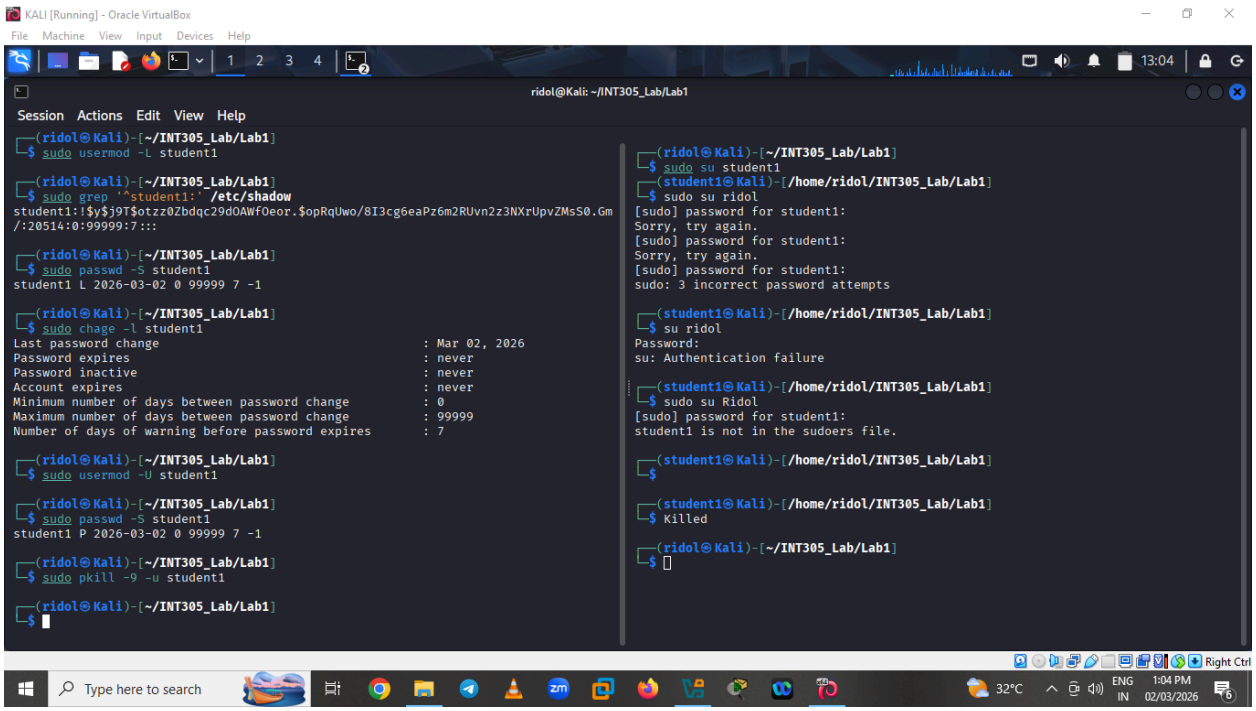

Fig 2 : Evidence showing successfully termination of user session

3.3 Locking User Account

- Command : **`sudo usermod -L student1`**
- Purpose : Lock the user account (prevents login)

- Command : **`sudo passwd -S student1`**
- Command : **`sudo grep '^student1:' /etc/shadow`**

- Purpose : Verify the account is locked ("L")

- Command **: `sudo usermod -U student1`**
- Purpose : Unlock the user account

- Command **: `sudo passwd -S student1`**
- **Purpose :** Verify the account is unlocked again ("P")

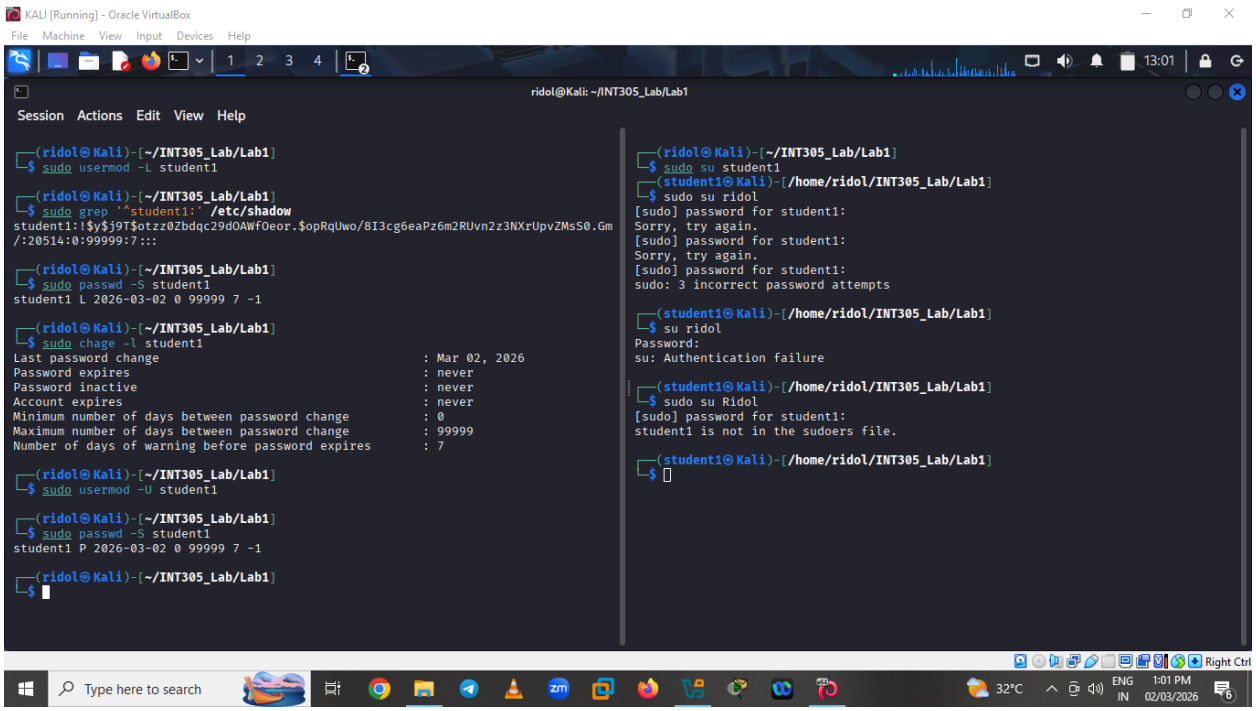

Fig 3 : Evidence showing successfully locked and unlocked user account

3.4 Deleting User account

- Command **: `sudo deluser student1`**
- Command **: `sudo userdel student1`**
- Purpose : Delete the user but keep the home directory

- Command **: `sudo deluser --remove-home student1`**
- Command **: `sudo userdel -r student1`**

- Purpose : Completely remove the user AND their home directory

- Command **: `id student1`**
- Purpose : Confirm the user is fully gone

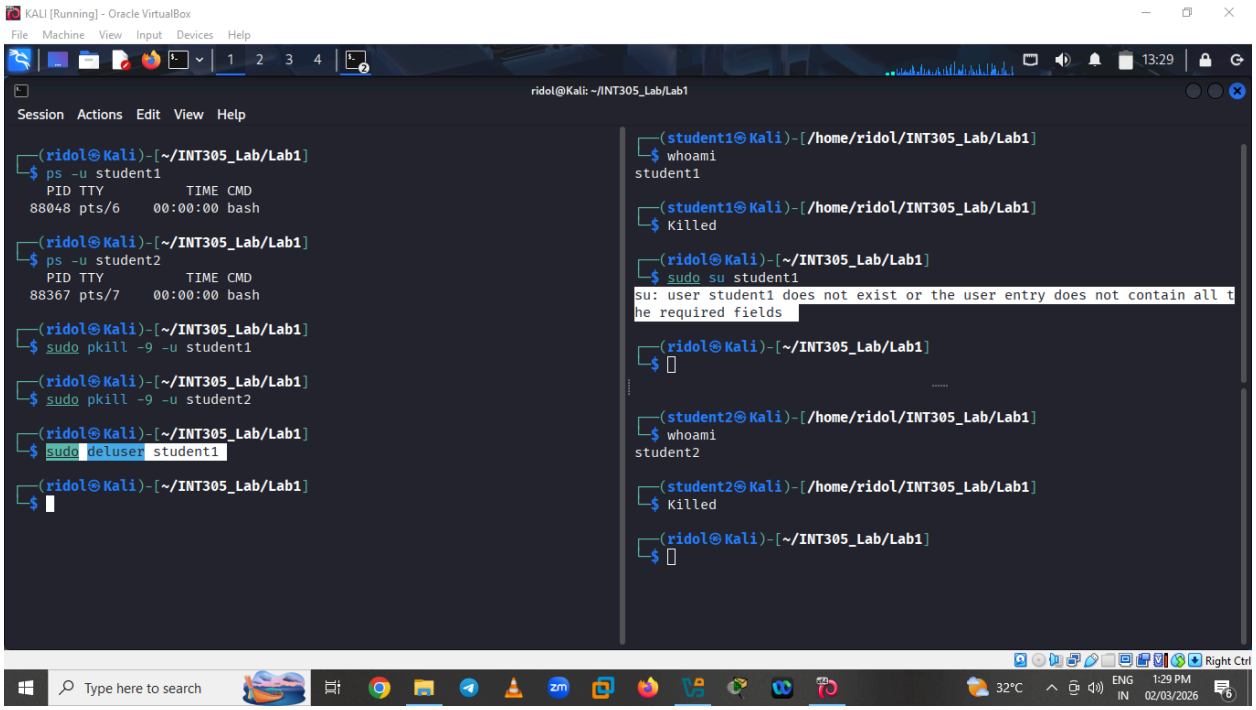

Fig 4: Evidence showing the user account being deleted

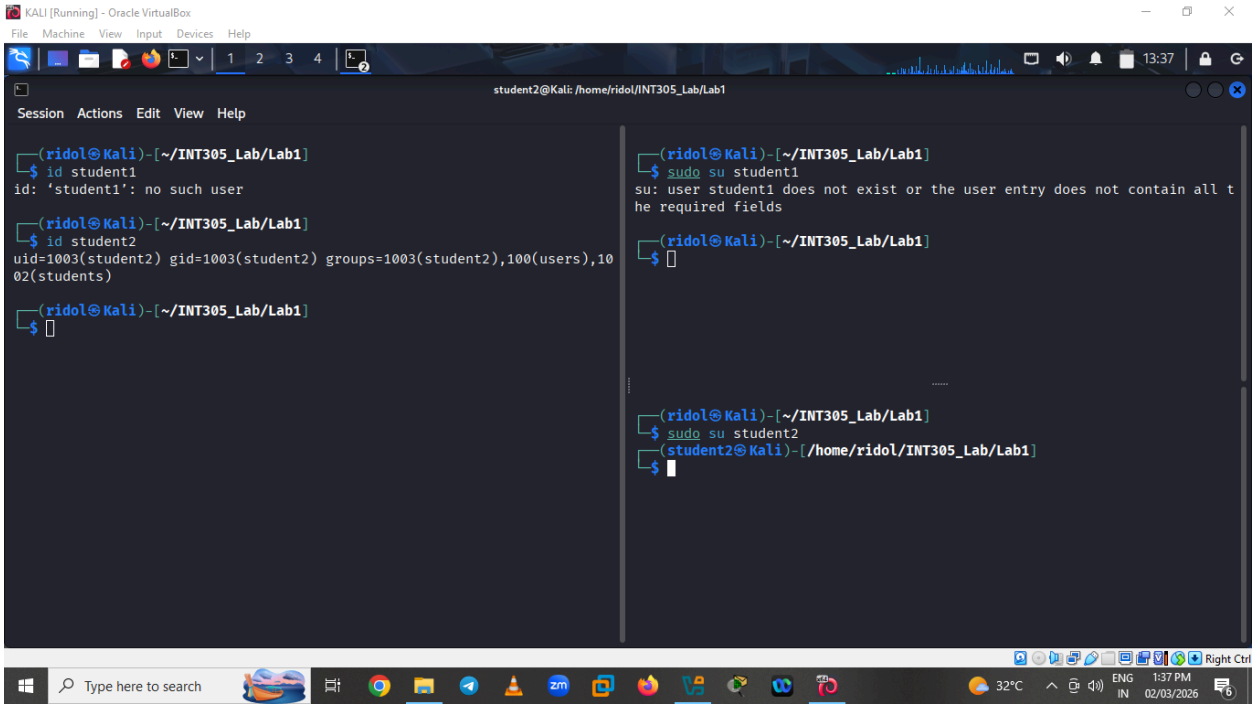

Fig 5: Evidence showing successful confirmation of the deleted user account

4.0 Results and Analysis

The added user "student1" has limited permission and does not have a super user privilege by default. A super user retains the permission and privileges to kill an active session of added users or perform access control on the user account being the administrator.

5.0 Conclusion and Recommendation

The lab justifiably elaborated the role and importance of user management by a system administrator. It is important that an administrator has complete control of user management for access control and incident response purposes in a case of security breach from a user as the attack vector.